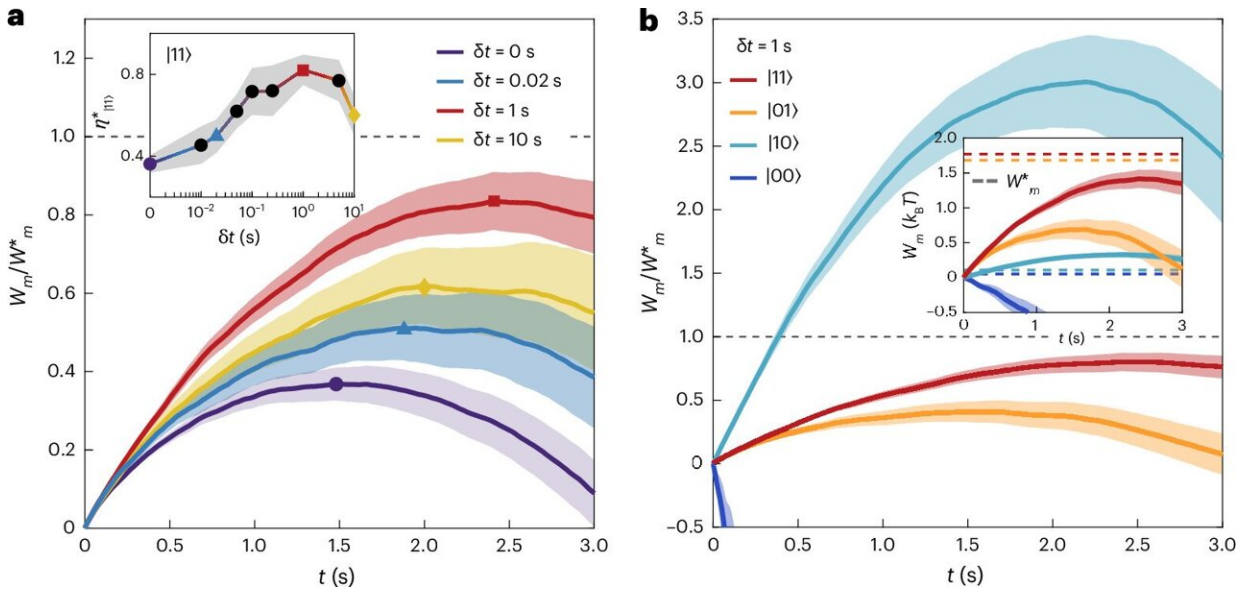
Maxwell's demon learns to turn concealed information into usable energy for microscopic systems

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Work extraction from hDoFs through two-measurement protocols. Credit: *Nature Physics* (2026). DOI: 10.1038/s41567-026-03442-w

Since its introduction more than 150 years ago, the thought-provoking idea of Maxwell's demon has profoundly shaped our understanding of the relationship between information and energy. Maxwell's demon is an imagined microscopic gatekeeper that exploits information at the microscopic scale to produce macroscopic work.

Initially controversial, the demon appeared to challenge a fundamental law of thermodynamics by suggesting that work could be extracted from a system in thermal equilibrium.

Today, this apparent paradox is resolved by recognizing the thermodynamic value of information: Information itself is a physical resource that can be converted into work. This insight has inspired growing interest in information engines—microscopic machines that, unlike conventional heat engines, are powered by information rather than by temperature differences.

The work is published in *Nature Physics*.

Channeling motion by means of information

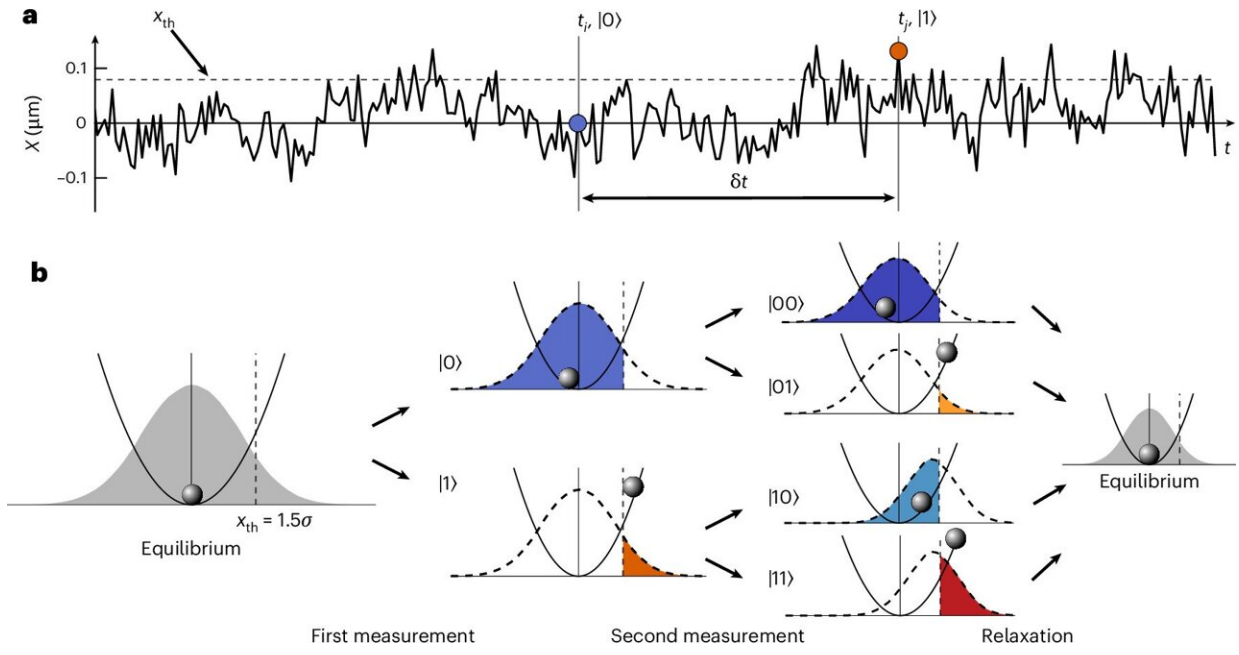
To understand how such a demon extracts work from information, consider a small door connecting two compartments filled with gas molecules. By monitoring the motion of individual molecules, the demon selectively opens the door for fast molecules moving in one direction and slow molecules moving in the other.

Over time, one compartment becomes hotter while the other becomes colder, creating a temperature difference that can be used to drive a machine and perform work. The key point is that the demon creates this energetic imbalance not by supplying energy itself, but solely by exploiting information about the microscopic particle motion.

In realistic settings, however, a demon never has complete information. Even for a single particle, its position and velocity cannot be measured with perfect accuracy. Moreover, microscopic machines are often expected to operate not in simple liquids such as water, but in far more complex surroundings—including polymer solutions, biological fluids and other structured materials—with an internal molecular

microstructure that influences the particle's motion.

Because such internal microstructures are not directly accessible to the demon, part of the relevant information remains hidden. This missing information limits the efficiency of microscopic machines under such conditions.



Two-measurement protocol and state-conditioned particle distributions. Credit: *Nature Physics* (2026). DOI: 10.1038/s41567-026-03442-w

Eye-patched: The demon's perspective

In a combined experimental and theoretical study, physicists at the University of Konstanz and the Max Planck Institute for Dynamics and Self-Organization in Göttingen have now shown that such hidden information can be exploited as a valuable thermodynamic resource.

In the experiments, a microscopic particle is confined by a tightly focused laser beam inside a complex liquid. Unlike a simple fluid, this medium forms a dynamic, spaghetti-like network of microscopic structures that continuously rearrange and interact with the particle.

As the particle moves, it deforms this network, which in turn exerts forces back on the particle and influences its later motion. While the particle itself can be tracked with high precision, the internal dynamics of the surrounding network cannot be observed directly.

From the demon's perspective, the environment is therefore only partially visible: Some of the information relevant for optimal control remains concealed in the medium. Without access to this hidden information, the demon's actions cannot be perfectly aligned with the state of the system, reducing the amount of work that can be extracted.

Double-measurement strategy reveals hidden information

To partially recover this missing information, the researchers use a double-measurement strategy. Instead of relying on a single observation, they measure the particle's position twice, separated by a controlled time delay.

By comparing two measurements separated in time, the researchers obtain indirect information about the hidden state of the surrounding medium. In this way, the demon gains indirect access to otherwise hidden environmental dynamics, extending its knowledge beyond a single snapshot in time.

By carefully tuning the delay between the two measurements, the researchers show that energy stored in the hidden dynamics of the

environment can be harvested and converted into useful work. Under suitable conditions, exploiting this hidden information allows the demon to extract more work than would be possible from a single measurement.

The results demonstrate that information encoded in hidden environmental degrees of freedom can be partially recovered and exploited as a thermodynamic resource. By exploiting temporal correlations and environmental memory, it becomes possible to harness energy from degrees of freedom that remain inaccessible to direct observation.

More broadly, the current study represents an important step toward cyclic microscopic information engines that, unlike conventional heat engines, can operate without requiring a temperature difference or other macroscopic energy gradient. This is particularly relevant at microscopic length scales, where maintaining stable thermal gradients becomes increasingly difficult.

Instead, such engines would use information about microscopic fluctuations as a genuine thermodynamic resource.

More information: The extraction of work from hidden degrees of freedom, *Nature Physics* (2026). DOI: 10.1038/s41567-026-03442-w. www.nature.com/articles/s41567-026-03442-w

Provided by University of Konstanz

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